

HONORS CHEMISTRY

INVESTIGATION 6:

Chemical Bonding

Theenash Sengupta, MSJ STEM Success - 10/8/2025

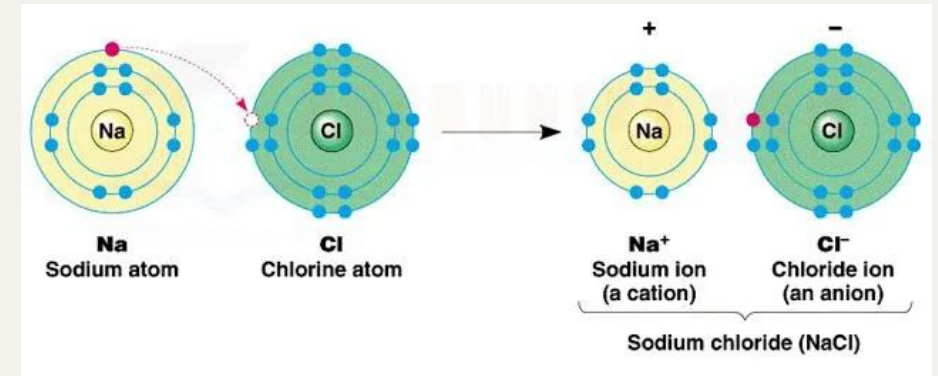




Follow Along!

Ionic Bonding

- Electrons are **transferred** from one atom to another. Two benefits:
 - Each atom has an **octet** electron config. (e.g. Na^+ has config. $[\text{Ne}]$, Cl^- has config. $[\text{Ar}]$)
 - Since the atoms are opposite charges, they are stabilized by **electrostatic interactions**.
- Properties of Ionic Solids:
 - Conduct electricity **when dissolved in water**
 - Relatively high melting points



Ionic Nomenclature

- Ionic bonds generally form between **metals** and **nonmetals**.
 - If the metal in the compound is in group 1 or 2 on the Periodic Table, you should be able to predict its oxidation state (Group 1 = +1, Group 2 = +2)
 - The charge on the nonmetal is generally 8-group #: e.g. O is in group 6, so its charge is 2-.
 - **Mg is Magnesium, Mn is Manganese!!!**
 - **Silver has only one oxidation state: +1**

1	2	3	4	5	6	7	8
H	s-block		p-block				He
Li	Be	B	C	N	O	F	Ne
Na	Mg	Al	Si	P	S	Cl	Ar
K	Ca	Ga	Ge	As	Se	Br	Kr
Rb	Sr	In	Sn	Sb	Te	I	Xe
Cs	Ba	Tl	Pb	Bi			

Practice

- Provide the chemical formula for each of the following ionic compounds:
 - Sodium Oxide
 - Aluminum Sulfide
 - Iron (II) Nitride
- Supply the name for each of the following ionic compounds based on the formula:
 - BeBr_2
 - Fe_2O_3
 - CaF_2

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Li	Be	B	C	N	O	F	Ne
Na	Mg	Al	Si	P	S	Cl	Ar
K	Ca	Ga	Ge	As	Se	Br	Kr
Rb	Sr	In	Sn	Sb	Te	I	Xe
Cs	Ba	Tl	Pb	Bi			

Practice Answers

- Provide the chemical formula for each of the following ionic compounds:
 - Sodium Oxide (Na_2O)
 - Aluminum Sulfide (Al_2S_3)
 - Iron (II) Nitride (Fe_3N_2)
- Supply the name for each of the following ionic compounds based on the formula:
 - BeBr_2 (Beryllium Bromide)
 - Fe_2O_3 (Iron (III) Oxide)
 - CaF_2 (Calcium Fluoride)

1	2	3	4	5	6	7	8
H	s-block		p-block				He
Li	Be	B	C	N	O	F	Ne
Na	Mg	Al	Si	P	S	Cl	Ar
K	Ca	Ga	Ge	As	Se	Br	Kr
Rb	Sr	In	Sn	Sb	Te	I	Xe
Cs	Ba	Tl	Pb	Bi			

Polyatomic Ions

- Focus on **Oxyanions**
- Rules:
 - Sulfate, Sulfite = SO_4^{2-} , SO_3^{2-} (ate > ite)
 - Carbonate = CO_3^{2-} , **bicarbonate = HCO_3^-**
 - Nitrate, Nitrite = NO_3^- , NO_2^-
 - Phosphate, Phosphite = PO_4^{3-} , PO_3^{3-} (phos fat)
 - Cl, Br, I $\rightarrow \text{XO}^-$, XO_2^- , XO_3^- , XO_4^- = hypo_ite, _ite, _ate, per_ate (hypo = less, per = more)
- Other ones you might need to know:
 - Peroxide = O_2^{2-} (e.g. sodium peroxide is Na_2O_2)
 - Acetate = CH_3COO^- (this was on the test last year!)
 - **Only polyatomic cation is NH_4^+ (ammonium)**

Practice

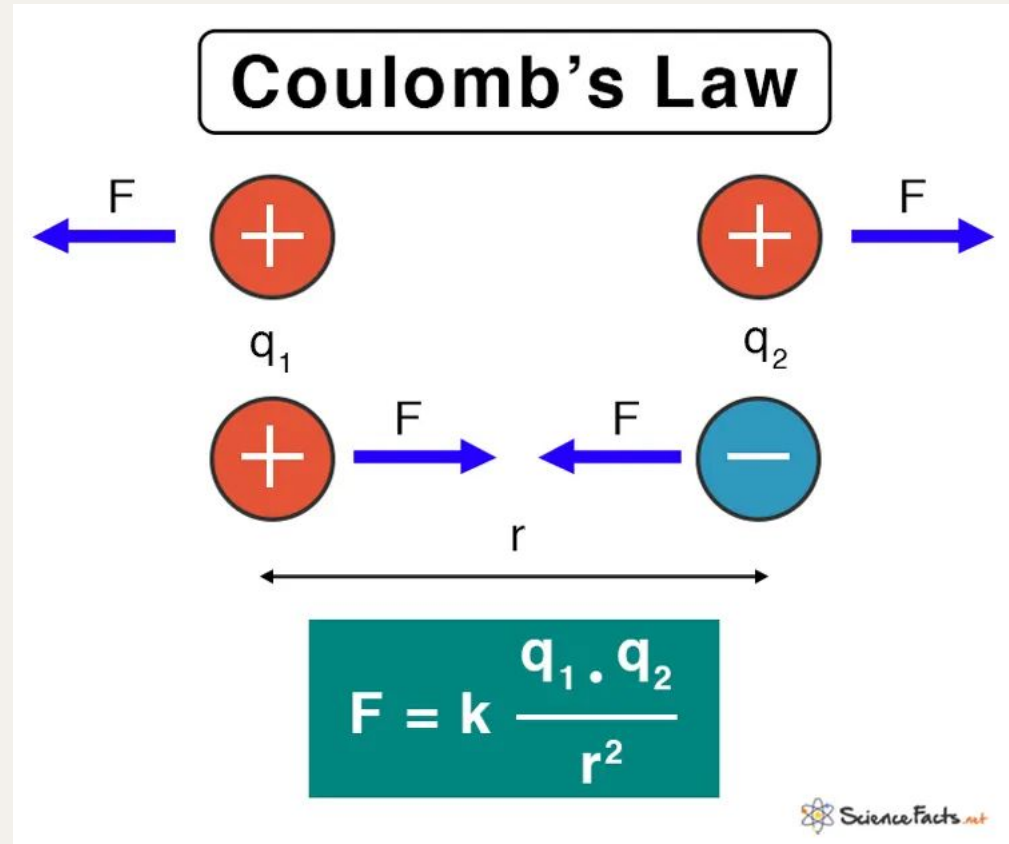
- Identify the chemical formula:
 - a. Silver Phosphate
 - b. Manganese (II) Nitrite
 - c. Barium Perchlorate
 - d. Rubidium Sulfate
 - e. Lithium Carbonate
- Find the name:
 - a. Na_3PO_3
 - b. $\text{V}(\text{NO}_3)_3$
 - c. $\text{Ca}(\text{ClO}_2)_2$

Practice

- Identify the chemical formula:
 - a. Silver Phosphate (Ag_3PO_4)
 - b. Manganese (II) Nitrite ($\text{Mn}(\text{NO}_2)_2$)
 - c. Barium Perchlorate ($\text{Ba}(\text{ClO}_4)_2$)
 - d. Rubidium Sulfate (Rb_2SO_4)
 - e. Lithium Carbonate (Li_2CO_3)
- Find the name:
 - a. Na_3PO_3 (Sodium Phosphite)
 - b. $\text{V}(\text{NO}_3)_3$ (Vanadium (III) Nitrate)
 - c. $\text{Ca}(\text{ClO}_2)_2$ (Calcium Chlorite)

Coulomb's Law

- Stability of an ionic compound is dependent on the charges of the constituent atoms and their respective radii. (don't ignore the distance term!)
- Strength of electrostatic interaction (force) is **directly** proportional to charge and **inversely** proportional to the square of distance (sum of ionic radii) between atoms.
- MgO and CsI are illustrative examples:
 - CsI is relatively **unstable** because the product of its charges is 1 and d is very large.
 - MgO is one of the **most stable** ionic solids because the product of its charges is 4 and d is very small.



Practice

Rank the following ionic solids by **increasing** stability based on Coulomb's law:

- 1) NaCl
- 2) NaI
- 3) FeS

Practice Answers

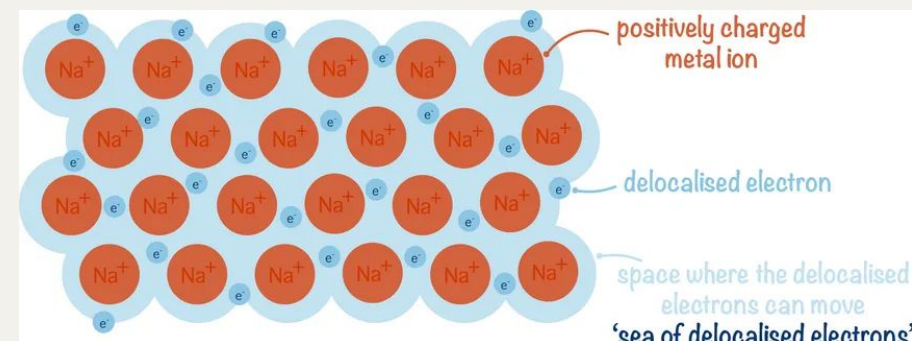
Rank the following ionic solids by **increasing** stability based on Coulomb's law:

- 1) NaCl
- 2) NaI
- 3) FeS

Answer: 2, 1, 3

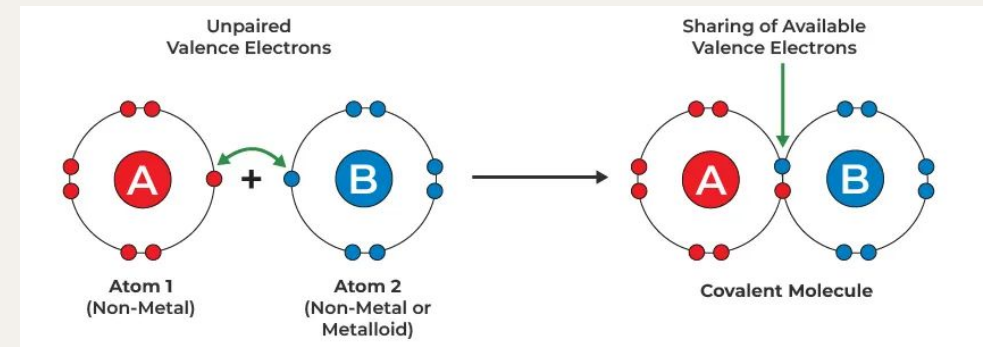
Metallic Bonding

- Sea of electrons model, fully delocalized electrons from all atoms in the solid holds it together
- Confers unique properties:
 - Relatively malleable (sheets) and ductile (wires)
 - Lower melting points than ionic solids, sodium melts more easily than magnesium



Covalent Bonding

- Sharing of electrons between atoms, usually between nonmetals
- Stabilized due to **octet** electron configuration of atoms
 - Hydrogen needs 2 electrons to be stable
- Visual descriptions of covalent bonding in a molecule can be shown through **Lewis structures**.



Nomenclature

- Covalent nomenclature rules:
 - *Number prefix* [element 1] *number prefix* [element 2]
 - Order of elements is left to right on periodic table
 - If element 1 appears once in the formula, no prefix
 - Ex. Phosphorus pentafluoride, dicarbon hexahydride
- Some common names:
 - Ozone (O_3)
 - Methane (CH_4)
 - Nitric Oxide (NO) (alternate name for nitrogen monoxide)

Practice

Name:

- a. CO
- b. N₂O₃
- c. SF₆
- d. PCl₃
- e. Cl₂O₇

Formula:

- a. Dinitrogen tetroxide
- b. Carbon disulfide
- c. Sulfur trioxide
- d. Phosphorus pentafluoride
- e. Iodine monochloride

Practice Answers

Name:

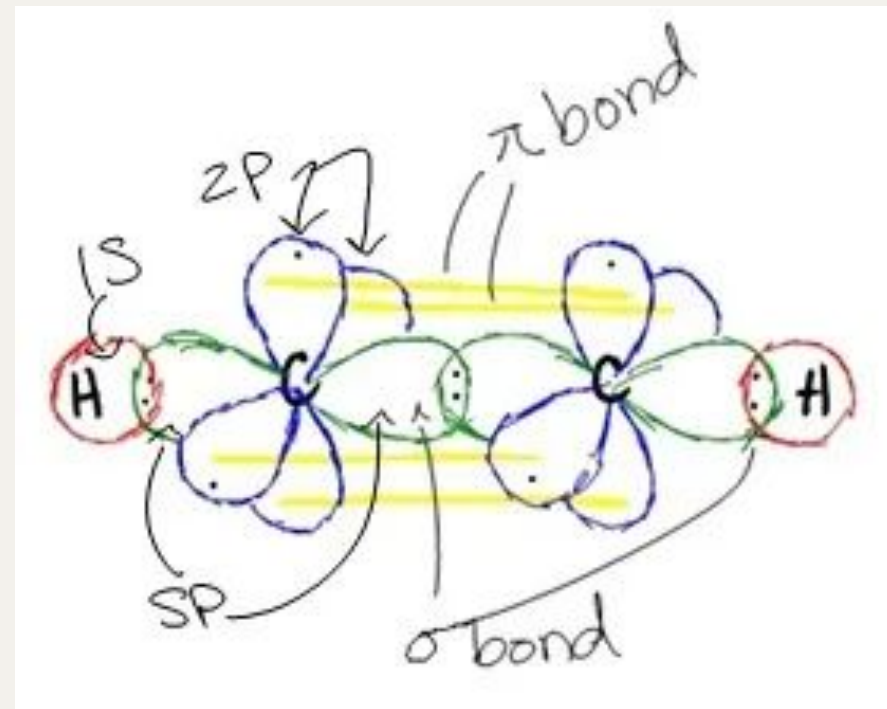
- a. CO (carbon monoxide)
- b. N_2O_3 (dinitrogen trioxide)
- c. SF_6 (sulfur hexafluoride)
- d. PCl_3 (phosphorus trichloride)
- e. Cl_2O_7 (dichlorine heptoxide)

Formula:

- a. Dinitrogen tetroxide (N_2O_4)
- b. Carbon disulfide (CS_2)
- c. Sulfur trioxide (SO_3)
- d. Phosphorus pentafluoride (PF_5)
- e. Iodine monochloride (ICI)

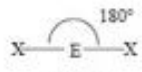
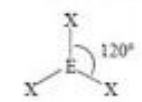
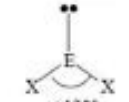
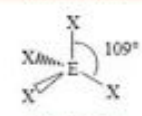
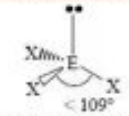
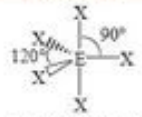
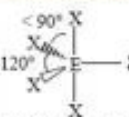
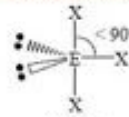
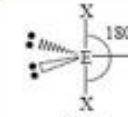
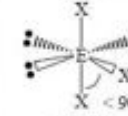
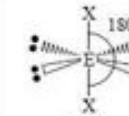
Hybridization

- Electron orbitals (s, p, d) mix to create hybrid orbitals that form more stable structures, shapes agree with VSEPR theory (next).
 - sp^3 : tetrahedral electron geometry
 - sp^2 : trigonal planar electron geometry
 - sp : linear electron geometry
- In sp^2 and sp hybridization, the other p orbitals are put toward forming **double** and **triple** bonds. (increasing # of bonds between 2 atoms = higher strength)



VSEPR

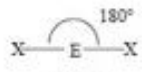
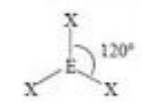
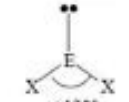
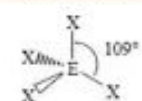
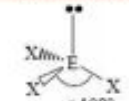
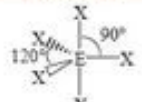
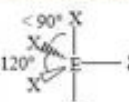
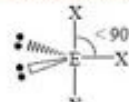
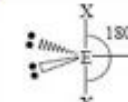
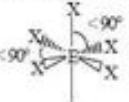
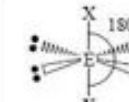
- Table at right →
- Can rationalize / derive by analyzing repulsions: choose the structure that minimizes repulsions between bonded substituents
 - Ex. T-shaped vs. trigonal pyramidal: lone pairs in the equatorial plane are further away from bonded groups (120° vs. 90°)
- If given a molecule, draw out the Lewis structure and deduce which structure the molecule best fits

Bonding Domain or Electron Domain	VSEPR Geometries				
	Basic Geometry 0 lone pair	1 lone pair	2 lone pairs	3 lone pairs	4 lone pairs
2	 Linear				
3	 Trigonal Planar	 Bent or Angular			
4	 Tetrahedral	 Trigonal Pyramidal	 Bent or Angular		
5	 Trigonal Bipyramidal	 Seesaw or Sawhorse	 T-shaped	 Linear	
6	 Octahedral	 Square Pyramidal	 Square Planar	 T-shaped	 Linear

VSEPR Practice

Identify the **molecular** geometry of each of the following molecules:

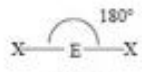
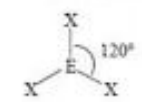
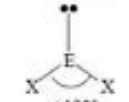
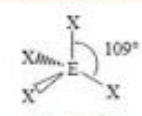
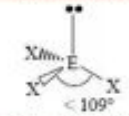
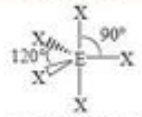
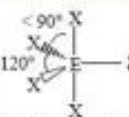
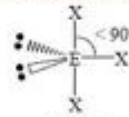
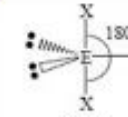
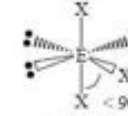
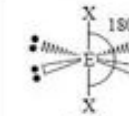
- XeF_4
- SF_6
- NH_3
- H_2O
- BCl_3
- BrF_5

Bonding Domain or Electron Domain	VSEPR Geometries				
	Basic Geometry 0 lone pair	1 lone pair	2 lone pairs	3 lone pairs	4 lone pairs
2	 Linear				
3	 Trigonal Planar	 Bent or Angular			
4	 Tetrahedral	 Trigonal Pyramidal	 Bent or Angular		
5	 Trigonal Bipyramidal	 Seesaw or Sawhorse	 T-shaped	 Linear	
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VSEPR Practice

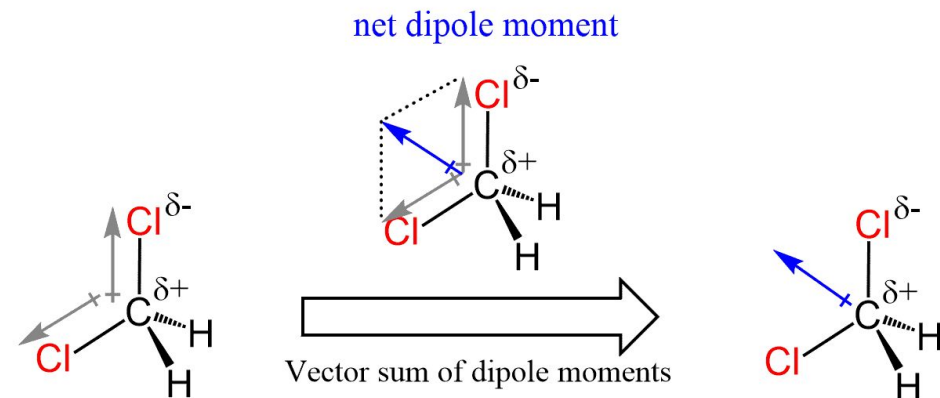
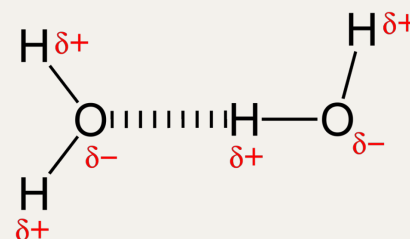
Identify the **molecular** geometry of each of the following molecules:

- XeF_4 (Square Planar)
- SF_6 (Octahedral)
- NH_3 (Trigonal Pyramidal)
- H_2O (Bent)
- BCl_3 (Trigonal Planar)
- BrF_5 (Square Pyramidal)

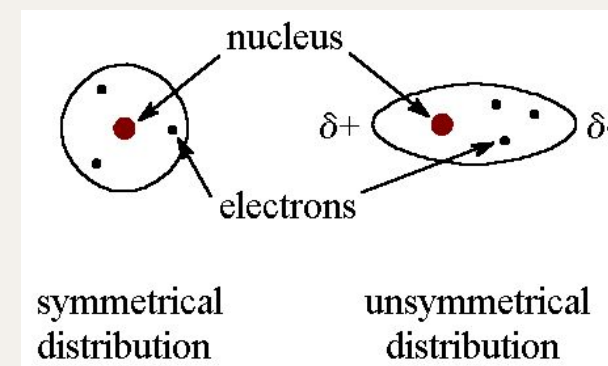
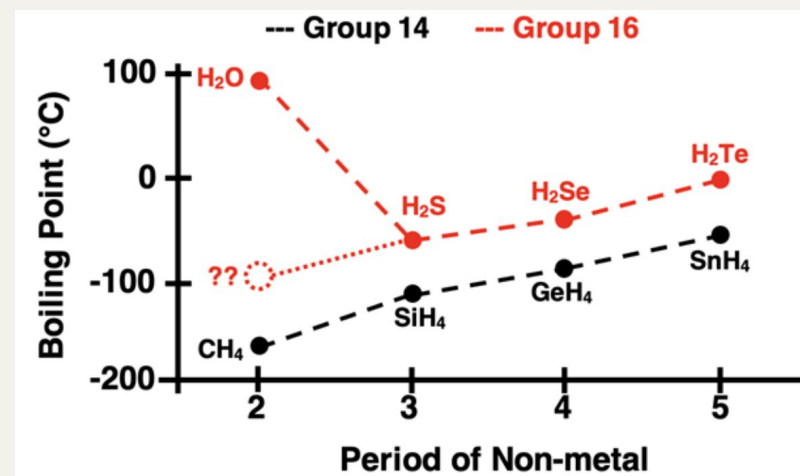
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Intermolecular Forces

- Molecules can be **polar** when there is an electronegativity difference between atoms, and the directions of the charge separation does not cancel.
 - Since polar molecules tend to attract each other, this leads to **dipole-dipole forces**, reducing the **volatility** of polar molecules.
- Highly electronegative elements (**Notice Of Failure**), when bonded to hydrogen, can share protons (hydrogen nuclei) between each other, resulting in **hydrogen bonding**.
 - This also reduces volatility significantly, e.g. water, ammonia
- Finally, electron cloud distortion promotes interactions between nonpolar molecules known as **London Dispersion Forces**
 - Larger atoms / bigger molecules tend to have stronger LDFs and are therefore less volatile.



Connect the vectors on the same starting point to find the vector sum.



Good Luck!

Any questions or feedback? Contact:

msjstemsuccess@gmail.com

